

EXAMINATION ASSIGNMENT

Hotel & Vacation Rental Competitor Intelligence Tracker

MakeMyTrip | Booking Intelligence | Availability Tracking | Revenue Analysis

1. Objective

Build a continuously running hotel and vacation-rental competitor intelligence system. The system must track a selected property and at least 30 comparable properties on MakeMyTrip, store repeated snapshots of publicly observable booking information, and convert the collected data into useful competitor, demand, availability and revenue analysis.

2. Specific Examination Task

Select one regularly bookable villa, resort or hotel in Lonavala or Karjat as the primary property. For this examination, treat that property as the property being analysed.

- Identify 30 comparable competitor properties on MakeMyTrip. This is mandatory.
- Track 50 competitors if possible. This is recommended.
- Tracking more than 50 comparable properties is a bonus.
- Build the system so the competitor list can be expanded without changing the core application.

3. MakeMyTrip: Version 1

MakeMyTrip is the mandatory platform for Version 1. The extraction and tracking component should be developed using Python or another appropriate technology, with Python strongly recommended for the data pipeline.

- Search and identify the selected property.
- Search and identify comparable properties.
- Collect the available booking, pricing and property information.
- Repeat the checks over time.
- Store every useful observation in a database.
- Compare each new snapshot against previous snapshots.

4. Booking Combination Testing

The system must investigate how availability and pricing change under different booking combinations.

- 1 guest
- 2 guests
- 3 guests
- 4 guests
- Higher guest counts where supported
- Different room quantities
- Different check-in dates

- Different check-out dates
- One-night and multi-night stays
- Weekdays versus weekends
- Near-term versus future dates

Students should determine which combinations produce useful market signals and explain how their tracking logic handles those combinations.

5. Availability as a Sales Signal

The system must store availability snapshots and compare them over time.

Example: if a property shows 8 rooms available in the morning and 5 rooms available during an evening check for the same booking conditions, the system records a decrease of 3 rooms and may treat that change as an estimated 3 room sales, subject to the assumptions documented by the student.

Students must distinguish observed availability changes from confirmed sales. A reduction in publicly displayed availability is a sales/demand signal, not direct access to the hotel's private booking ledger.

6. Data to Capture

Capture as much publicly available information as practical, including:

- Property name
- Property URL
- Location
- Room types
- Number of rooms or available units, where observable
- Available rooms/units
- Price
- Discounts
- Taxes and fees where available
- Guest capacity
- Number of guests selected
- Room quantity selected
- Check-in/check-out dates
- Stay duration
- Meal/breakfast/buffet options
- Cancellation conditions
- Amenities
- Rating
- Review count
- Other publicly visible demand or booking signals

7. Historical Database

Every observation must be stored rather than overwritten. The database should allow the system to be reviewed after one month or longer.

- Property
- Timestamp
- Search/booking conditions
- Availability
- Price
- Room type
- Guest count
- Stay dates
- Observed changes
- Source information

8. Change Detection

The system should compare the current observation with previous observations and identify meaningful changes.

- Rooms/units decreased
- Rooms/units increased
- Property became sold out
- Price increased
- Price decreased
- Discount changed
- Room type disappeared or became available
- Meal/buffet option changed
- Booking conditions changed
- Rating/review information changed

9. Estimated Booking & Revenue Model

Build an estimation model using the collected snapshots. The model should estimate booking activity, room nights and revenue based on observed changes.

At minimum, the analysis should consider:

- Estimated rooms/units sold
- Estimated room nights sold
- Average observed selling price
- Estimated gross booking value/revenue
- Occupancy or availability trend
- Weekday versus weekend performance
- Property-level booking velocity

All estimated figures must clearly state the assumptions and methodology used. Students must not present estimated revenue as confirmed private revenue.

10. Competitor Intelligence

The purpose of the database is to understand what is happening across the selected market.

- Which properties are showing the largest availability reductions?
- Which properties appear to be selling faster?
- Which properties are frequently sold out?
- Which properties increase prices as dates approach?
- Which properties have stronger weekend pricing?
- Which properties offer discounts or meal options?
- How does the selected property compare with its competitors?
- How does the market change over time?

11. Dashboard

Create a dashboard that turns the collected database into actionable analysis.

- Selected property versus competitors
- 30+ competitor overview
- Price comparison
- Availability comparison
- Estimated rooms sold
- Estimated revenue
- Estimated occupancy
- Booking velocity
- Sold-out frequency
- Weekday versus weekend comparison
- Guest-count comparison
- Room-type comparison
- Historical trends
- Top changes since previous check

12. Tracking Frequency

Students may choose their tracking frequency based on the system architecture and available resources. They must explain the chosen frequency and why it is appropriate for the analysis.

The system must support repeated tracking and preserve historical snapshots so that performance can be analysed after weeks or months.

13. Scale Testing

The system should be designed so that adding more competitors does not require redesigning the application.

- Demonstrate the mandatory 30 competitors.
- Attempt 50 competitors where possible.
- Explain how the system would handle 100 or more properties.
- Use background processing, queues, batching or concurrency where appropriate.
- Prevent one failed property from stopping the complete tracking process.

14. Live Data Demonstration

The system must be demonstrated using live MakeMyTrip data. Students should select properties and booking dates where availability and pricing can be observed and tracked over time.

The demonstration should show the same property being checked repeatedly and the resulting changes being stored in the database.

15. Version 2: Booking.com

Booking.com is an additional challenge. Students may attempt to extend the same architecture to Booking.com.

- Investigate how its hotel search and availability data works.
- Determine what publicly accessible information can legitimately be collected.
- Adapt the data model where required.
- Demonstrate the same competitor analysis if successful.

Students must respect applicable access controls, terms and permissions. Security controls should not be bypassed.

16. Bonus: Other Platforms

Students may attempt additional booking platforms such as Agoda, Goibibo, Airbnb, Expedia, Cleartrip or other relevant platforms. Successfully extending the architecture to another platform is a bonus.

17. Technical Requirements

- Python is recommended for the extraction and tracking engine.
- Use a persistent database.
- Do not keep the complete dataset only in memory.
- Implement duplicate prevention.
- Implement error handling and retry logic.
- Keep historical snapshots.
- Separate raw collected data from calculated metrics where practical.
- Design the system so additional competitors can be added easily.

18. Mandatory Demonstration

- Show the selected Lonavala or Karjat property.
- Show at least 30 competitors.
- Demonstrate different guest and booking combinations.
- Show repeated MakeMyTrip tracking.
- Show availability changes being detected.
- Show price changes being detected.
- Show data being stored historically.
- Show the estimated booking/revenue calculations.
- Show competitor comparison in the dashboard.
- Explain how the system scales to 50+ properties.

19. Required Deliverables

- Complete source code
- Working tracking application
- Database and schema
- 30+ competitor dataset
- Dashboard
- Revenue/booking estimation methodology
- Technical architecture documentation
- Live-data demonstration
- Demonstration video
- Instructions to run and test the system

20. Bonus Challenges

- 50+ competitors
- More than 50 competitors
- Booking.com integration
- Additional booking platforms
- Long-term automated tracking
- Advanced demand prediction
- More detailed revenue modelling
- Automated competitor discovery
- Large-scale distributed tracking
- Advanced market trend analysis

21. Expected Outcome

The finished system should provide a historical intelligence database for the selected hotel/villa market. After sufficient tracking, the user should be able to understand how availability, pricing and observable booking signals are changing across the selected property and its competitors, and use those observations to estimate market performance and revenue opportunity.

Examination Focus: Data Collection • Automation • Historical Tracking • Analysis • Revenue Modelling